Dongyoung Kim, Ph.D.

Artificial intelligence for science and engineering

Date of birth: 1987.05.26

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EXPERIENCE

2017 – Current	**Institute for Basic Science**, South Korea Research Fellow, Advisor: Prof. Yoon-Kyung Cho (Ulsan National Institute of Science and Technology).
2012 – 2016	**Texas A&M University,** U.S.A. Research Assistant, Advisor: Raimund Ober (Texas A&M University).
2010 – 2012	**The University of Southwestern medical center**, U.S.A. Research Assistant,
2010 – 2012	**The University of Texas at Dallas**, U.S.A. Research Assistant
2010 – 2010	**EOS Australia,** Sydney, Australia Intern
2009 – 2010	**Embedded software research center,** South Korea Research Assistant

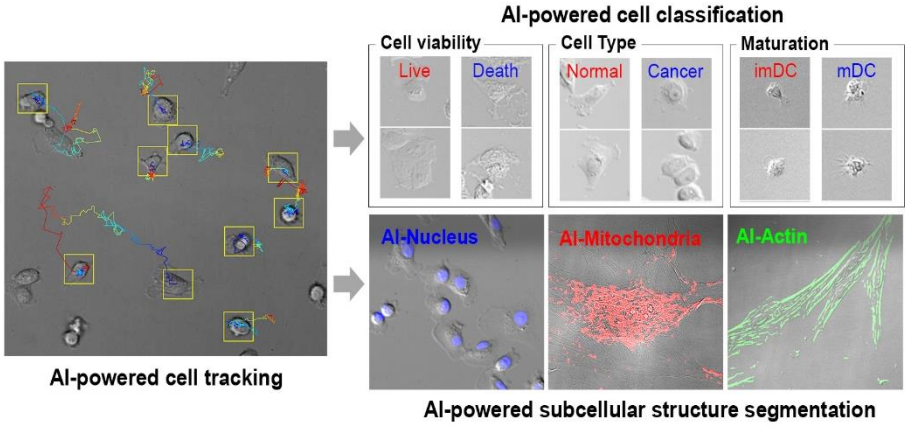
EDUCATION

2012 – 2016	**Ph.D., Biomedical Engineering,** **Texas A&M University** College station, Texas, USA
2010 – 2012	**B.S., Electrical engineering and computer science** **The University of Texas at Dallas** Richardson, Texas, USA
2006 – 2012	**B.S. Electrical engineering and computer science** **Kyungpook National University** Daegu, South Korea

PROJECTS

2017– 2019

AI-powered transmitted light microscopy for functional analysis of live cells.



- Transforming a transmitted light microscopy image into a fluorescence microscopy equivalent image (or information) using unsupervised machine learning and supervised semantic segmentation.
- Real-time monitoring of living cell property and status from transmitted light microscopy video using region proposal convolutional neural network and convolutional neural network.
- Living cell tracking in time-lapse video using complementary learning of object shape and colors online.

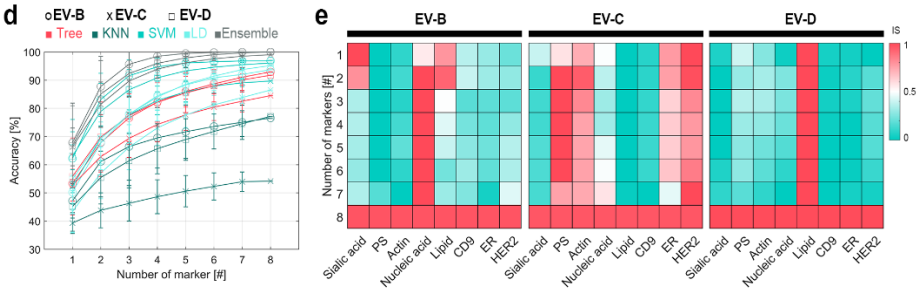
*Two Korea patents filed (2019).

*Software developed using **MATLAB**

*Manuscript under review (Sci. Rep. 2019).

2017 – 2019

Find cancer-related particles from blood using machine learning



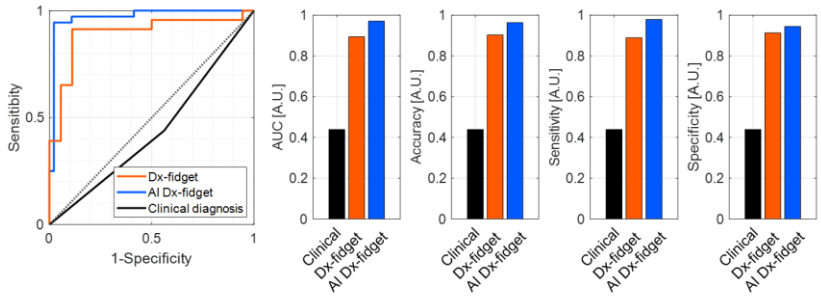
- Cancer prediction from blood born particle using machine learning classification.

*Software developed using **MATLAB**

*Manuscript under review (Anal. Chem. 2019).

2017 – 2019

Point-of-care-testing urinary tract infection diagnosis using deep neural network



- Urinary tract infection diagnosis from a cell phone photograph of a testing kit using convolutional neural network.

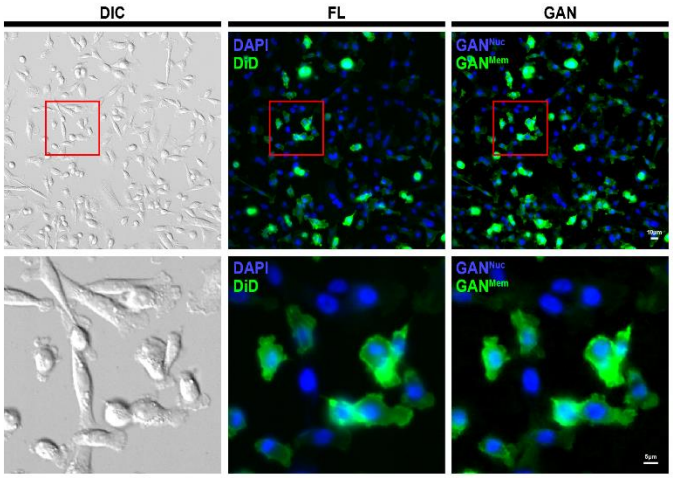
*A Korea patents filed (2018).

*Software developed using **MATLAB**

*Manuscript under review (Nature BME. 2019).

2019 – Current

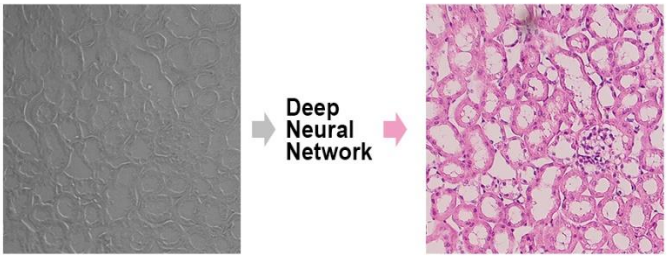
In-silico fluorescence staining using generative adversarial network.



*Manuscript under preparation (2019).

2019 – Current

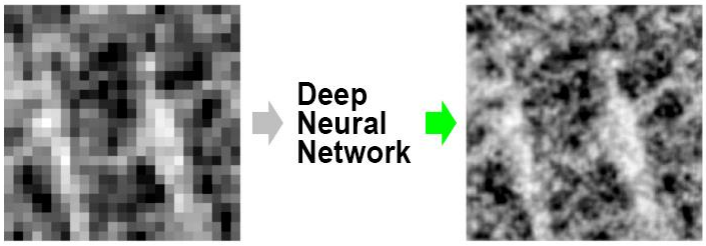
In-silico histology staining using generative adversarial network.



*Manuscript under preparation (2019).

2019 – Current

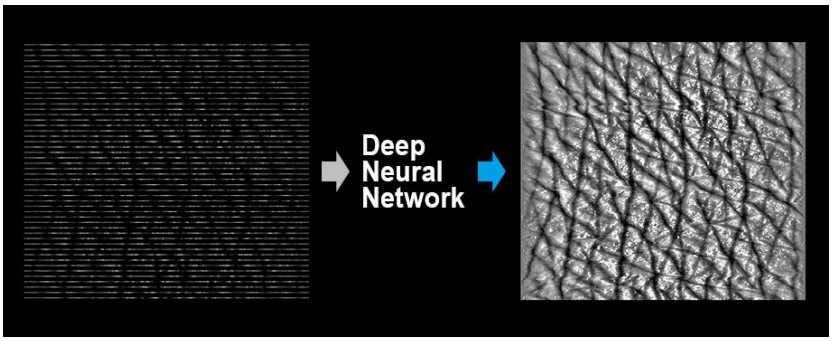
Generative adversarial network for high definition medical image recovery.



*Manuscript under preparation (2019).

2019 – Current

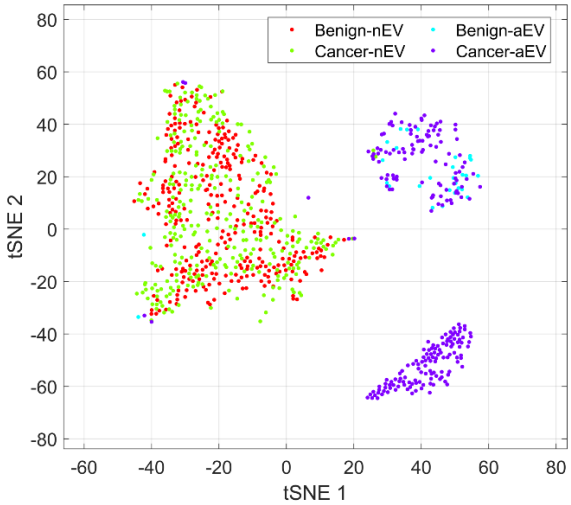
Generative adversarial network for rapid medical data acquisition.



*Manuscript under preparation (2019).

2019 – Current

Searching extreme rare cancer extracellular vesicle in blood plasma using generative adversarial network.



*Manuscript under preparation (2019).

PATENTS

Kim D., Min Y., Cho Y., **Analyzing method for cell image using artificial neural network and image processing apparatus for cell image**. Patent filed #10-2019-0001740, South Korea, 2019.

Kim D., Min Y., Cho Y., **Generating method for specialized microscope images using artificial neural network and image processing apparatus**. Patent filed #10-2019-0001741, South Korea, 2019.

Kim D., Micheal I., Ki D., Cho Y., Centrifugal force based non-powered particle filtration apparatus and method of particle filtration. Patent filed #10-2018-113085, South Korea, 2018.

Kim D., Ober R.J, **Advanced multi-dimensional microscopy system for single particle & structure imaging.** U.S. Patent pending, 2017.

HONORS & AWARDS

2018	1st place at Shark Tank Competition. **The 22nd International Conference on Miniaturized Systems For Chemistry And Life Sciences**
2016	1st place at 3D single molecule localization microscopy **SMLMS Challenge, EPFL, Switzerland.** *Software developed using **MATLAB**
2014 – 2016	Scholarship, **Texas A&M University, U.S.A.**
2012 – 2014	Scholarship, **The University of Texas at Dallas, U.S.A.**
2011 – 2011	Scholarship, **The University of Texas at Dallas , U.S.A.**
2011 – 2011	2nd place for Senior Design Project, **The University of Texas at Dallas**
2010 – 2012	Scholarship, **The University of Texas at Dallas, U.S.A.**

LANGUAGES

English, Korean

SKILLS

Artificial Intelligence	Supervised/unsupervised machine learning, Deep neural network design, generative adversarial network design.
Programming	MATLAB, Python, C, C++, JAVA.

Data science	Big data analysis, Optimization based data processing.
Electrical system architecture	NI LabWindows/CVI, NI LabView.
Optical system architecture	Zemax based optical system design, Multimodal optical system configuration.
Microscope	Epifluorescence, TIRF microscopy, Confocal microscopy Super-resolution microscopy Single molecule microscopy Transmitted/scanning electron microscopy

LECTURES

BME21101 Introduction to biomedical engineering (**Introduction to artificial intelligence for biomedical engineering / Introduction to deep learning**), 2019, Ulsan National Institute of Science and Technology (UNIST).

PUBLICATIONS

Kim. D., Min Y., Cho Y. **AI-powered transmitted light microscopy for functional analysis of live cells.** Scientific report 2019 (under review).

Kim. D., Woo H., Lee C,. Min Y., Cho Y. **Unveiling tumor specific extracellular vesicles by size fractionation and single vesicle analysis.** Analytical chemistry 2019 (under review).

Kim. D. Michael I**,** Oleksandra G., Kuimar S., Clara, J., Ki D., Cho Y. **Dx-Fidget Spinner: A Pocket-Sized Low-Cost Hand-Powered Medical Diagnostic Tool.** Nature Biomedical Engineering 2019 (under review).

Sunkara, V., Kim C., Park J., Woo H., **Kim D.**, Ha H., Kim M., Son Y., Kim J., Cho Y., **Fully automated, label-free isolation of extracellular vesicles from whole blood for cancer diagnosis and monitoring.** Theranostics, 2019.

Michael I., Kumar S., Oh J,, **Kim D**, Kim J., Cho Y. **Surface Engineered Paper Hanging Drop Chip for 3D Spheroid Culture and Analysis.** ACS applied materials & interfaces, 2018.

Li, R., Chiguru, S., Li, L., **Kim, D.**, Velmurugan, R., Kim, D., Tian, H., Schroit, A., Mason, R., Ober, R. J. and Ward, E. S. **Targeting phosphatidylserine with calcium-dependent protein-drug conjugates for the treatment of cancer.** Molecular Cancer Therapeutics, 2018.

Devanaboyina, S. C., Khare, P., Challa, **Kim, D.,** Ober, R. J., and Ward, E. S. **Engineered clearing agents for the selective depletion of antigen-specific antibodies.** Nature Communications, 2017.

Kim, D., Chao, J., Velmurugan, R., You, S., Ward, E. S., and Ober, R. J. **Remote focusing multifocal plane microscopy for the imaging of 3D single molecule dynamics with cellular context.** Proceedings of the SPIE, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXIV, 10070: 2017.

Vahid, M. R., Chao, J., **Kim, D.,** Ward, E. S., and Ober, R. J. **State space approach to single molecule localization in fluorescence microscopy.** Biomedical Optics Express, 2017

Poovasery J.S., Kang J.C., **Kim D**, Ober R.J, Ward E.S. **Antibody targeting of HER2/HER3 signaling overcomes heregulin-induced resistance to PI3K inhibition in prostate cancer.** International Journal of Cancer 2017

Kim D, You S, Ward E.S., Ober R.J, **Imaging of three-dimensional single molecule dynamics with cellular context: Antibody trafficking and interaction with cell membrane and sorting endosomes.** ASCB, San Fransisco, CA. 2016.

Cohen E. A. K.,. **Kim D**., Ober R.J, **Cramer-Rao Lower Bound for Point Based Image Registration with Heteroscedastic Error Model for Application in Single Molecule Microscopy.** IEEE Transactions on Medical Imaging 2015

Ram S, **Kim D**, Ober RJ, Ward ES. **The level of HER2 expression is a predictor of antibody-HER2 trafficking behavior in cancer cells**. mAbs 2014

Devanaboyina SC, Lynch SM, Ober RJ, Ram S, **Kim D**, Puig-Canto A, Breen S, Kasturirangan S, Fowler S, Peng L, et al. **The effect of pH dependence of antibody-antigen interactions on subcellular trafficking dynamics**. mAbs 2013

Ram S, **Kim D**, Ober RJ, Ward ES. **Microscopy in Polarized Epithelia Reveals a Novel Cellular Process of Intercellular Transfer.** Biophysical Journal 2013

Ram S, **Kim D**, Ober RJ, Ward ES. **3D Single Molecule Tracking with Multifocal Plane Microscopy Reveals Rapid Intercellular Transferrin Transport at Epithelial Cell Barriers**. Biophysical Journal 2012

Kim D, Ram S, Ober RJ, Ward ES. **3D single molecule tracking of rapid intracellular trafficking imaged by multifocal plane microscopy** Microscopy and Microanalysis 2012.